Here's a detailed comparison of **Kubernetes**, **AWS**, **Azure**, and **GCP** services grouped by category and shown in a **tabular format**, helping you map equivalent services across platforms for easier migration, understanding, or architecture planning.

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1. Orchestration and Compute Services

Category	Kubernetes	AWS	Azure	GCP	Use Case
Container Orchestration	Kubernetes (self-managed or via cloud)	Amazon EKS	Azure AKS	Google Kubernetes Engine (GKE)	Container orchestration, microservices
Virtual Machines	N/A	Amazon EC2	Azure Virtual Machines	Google Compute Engine	Traditional compute workloads
Serverless	Knative (on Kubernetes)	AWS Lambda	Azure Functions	Cloud Functions	Event-driven execution
Containers (Managed)	Kubernetes with CRI-O/Docker	AWS Fargate	Azure Container Instances (ACI)	Cloud Run	Run containers without managing servers

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2. Storage Services

Category	Kubernetes (via CSI)	AWS	Azure	GCP	Use Case
Block Storage	CSI + EBS or other plugin	Amazon EBS	Azure Disk Storage	Persistent Disk	Persistent volumes for pods
Object Storage	N/A	Amazon S3	Azure Blob Storage	Google Cloud Storage	Store unstructured data
File Storage	CSI + NFS plugin	Amazon EFS	Azure Files	Filestore	Shared file system
Backup & Snapshots	Velero or Kasten	AWS Backup	Azure Backup	Backup for GKE	Backup and disaster recovery

3. Networking Services

Category	Kubernetes (Core or CNI Plugin)	AWS	Azure	GCP	Use Case
Virtual Network	CNI + VPC	Amazon VPC	Azure Virtual Network	VPC	Private network for resources
Load Balancer	Service type=LoadBalancer	ELB/ALB/NLB	Azure Load Balancer / App GW	Cloud Load Balancing	Expose services to public/internet
Ingress Controller	NGINX/HAProxy/Traefik Ingress	ALB Ingress Controller	Application Gateway Ingress	Cloud Load Balancer Ingress	HTTP routing
DNS	CoreDNS + ExternalDNS	Route 53	Azure DNS	Cloud DNS	Domain resolution
Service Discovery	Kubernetes DNS	Cloud Map	Azure Service Discovery	Cloud Service Directory	Internal service name resolution

4. Database Services

Category	Kubernetes	AWS	Azure	GCP	Use Case
Relational DB	StatefulSet + MySQL/PostgreSQL	Amazon RDS / Aurora	Azure SQL Database / PostgreSQL	Cloud SQL / AlloyDB	Relational data
NoSQL DB	StatefulSet + MongoDB/Cassandra	DynamoDB	Azure Cosmos DB	Cloud Datastore / Firestore	Unstructured/ scalable data
Caching	Redis via Helm chart	Amazon ElastiCache	Azure Cache for Redis	Memorystore	Speed up app response time

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5. Security and IAM

Category	Kubernetes	AWS	Azure	GCP	Use Case
IAM	RBAC, OPA, Gatekeeper	AWS IAM	Azure AD + RBAC	IAM	Identity and access control
Secrets Management	Kubernetes Secrets / Vault	AWS Secrets Manager	Azure Key Vault	Secret Manager	Store sensitive config
Network Policies	Calico / Cilium	Security Groups / NACLs	NSGs / ASGs	VPC Firewall Rules	Traffic restrictions

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6. Monitoring, Logging & Observability

Category	Kubernetes	AWS	Azure	GCP	Use Case
Monitoring	Prometheus + Grafana	CloudWatch	Azure Monitor	Cloud Monitoring (Stackdriver)	Resource and app metrics
Logging	Fluentd / Loki / EFK Stack	CloudWatch Logs	Azure Log Analytics	Cloud Logging	Centralized log management
Tracing	Jaeger / OpenTelemetry	AWS X-Ray	Azure Application Insights	Cloud Trace	Distributed tracing
Alerting	Prometheus Alertmanager	CloudWatch Alarms	Azure Alerts	Cloud Monitoring Alerts	Set alarms on metrics

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7. CI/CD and DevOps

Category	Kubernetes	AWS	Azure	GCP	Use Case
CI/CD Tooling	ArgoCD, Tekton, Jenkins, Flux	CodePipeline / CodeBuild	Azure DevOps / GitHub Actions	Cloud Build / Cloud Deploy	Automate build/deploy

GitOps	ArgoCD / Flux	N/A	N/A	Config Sync (Anthos Config Mgmt)	Declarative deployment
Container Registry	Any registry (Harbor, etc.)	Amazon ECR	Azure Container Registry	Artifact Registry / Container Registry	Image storage

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8. AI/ML and Data Processing

Category	Kubernetes	AWS	Azure	GCP	Use Case
ML/AI Platform	Kubeflow	SageMaker	Azure Machine Learning	Vertex AI	Model training/deployment
Data Pipeline	Apache Airflow on K8s	AWS Glue / Data Pipeline	Azure Data Factory	Cloud Dataflow / Composer	ETL pipelines
Batch Processing	K8s CronJobs + Spark Operator	AWS Batch / EMR	Azure Batch	Dataproc / Dataflow	Big data processing

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9. Hybrid / Multi-cloud / Edge

Category	Kubernetes Native / Add-ons	AWS	Azure	GCP	Use Case
Multi-cloud Management	Rancher, Anthos, OpenShift	AWS Outposts, EKS Anywhere	Azure Arc	Anthos	Manage K8s clusters across clouds
Edge Compute	K3s, MicroK8s, EKS Distro	AWS Wavelength	Azure Stack Edge	GCP Edge Network	Low-latency workloads at edge

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Summary Table – Kubernetes Equivalent Across Clouds

Function	Kubernetes	AWS	Azure	GCP
Managed Kubernetes	Kubernetes	Amazon EKS	Azure AKS	Google GKE
Object Storage	N/A (via CSI/Sidecar)	Amazon S3	Azure Blob	Cloud Storage
Load Balancing	Ingress Controller	ELB / ALB	App Gateway / Load Bal.	Cloud Load Balancing
CI/CD	ArgoCD / Tekton	CodePipeline/Build	Azure DevOps	Cloud Build
Monitoring	Prometheus + Grafana	CloudWatch	Azure Monitor	Cloud Monitoring
Secrets Management	Kubernetes Secrets / Vault	Secrets Manager	Key Vault	Secret Manager
Container Registry	Harbor / DockerHub / etc.	ECR	ACR	Artifact Registry

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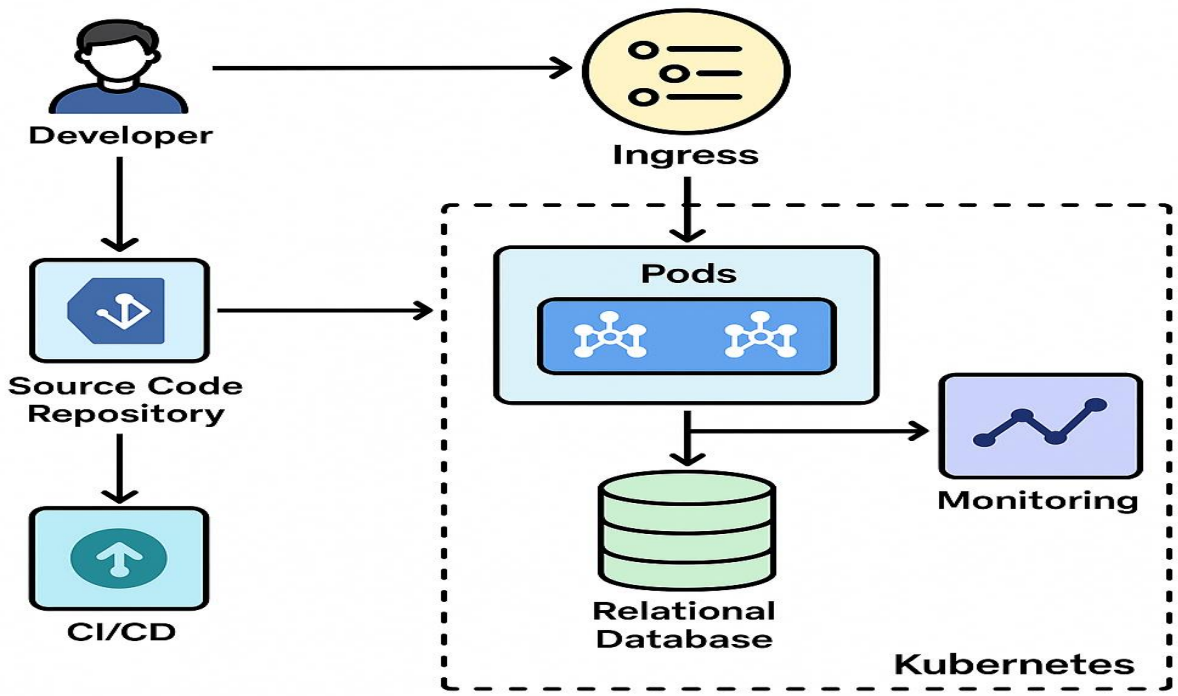
If you'd like architecture diagrams showing **how these services interact in real deployments (CI/CD + Monitoring + Ingress + DBs)**, let me know — I can generate visual diagrams too.


Developer
Ingress
Source Code
Repository
Pods
Monitoring
CI/CD
Relational
Database
Kubernetes